Tan Vi Hang

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EDUCATION

Beijing Institution Of Technology — China, Beijing

Bachelor in Computer Science And Technology — *Aug 2020 - Jul 2024*

CERTIFICATION

Coursera, Google — Online

Professional Certificate in UX Design — *2024*

Interaction Design Foundation (IxDF) — Online

AI for designer, Design Thinking — *2026*

EXPERIENCE

Lead Engineer — Dec 2025 - Present

OM Film Lab (Self Initiated Project) — *Malaysia, Remote*

- **Offline-First System Architecture:** Architected an offline-first data model using React Native (Expo), Zustand for client-side state, and local storage persistence. Allowed full app functionality—including raw color processing and preset editing—in zero-connectivity environments (e.g., outdoor shooting locations), while drastically eliminating unnecessary server round-trips and API costs.
- **Resilient Sync Engine:** Designed a robust background synchronization pipeline using React Query and WebDAV integration; automatically queues user modifications, preset data, and cloud backups locally, seamlessly executing bi-directional sync once high-speed network connection is restored without blocking the UI thread.
- **Proprietary UX/UI Innovation:** Designed and engineered a custom one-handed radial dial UI in React Native, compressing 30+ complex color parameters (HSL, WB, tone curves) into a single gesture-driven control—reducing user adjustment time from minutes to seconds.
- **Hardware & Wireless Integration:** Implemented OM System's Wi-Fi Direct transfer protocol, enabling seamless, cable-free camera-to-app photo syncing with automated EXIF metadata parsing and timeline organization.
- **Monetization & Complete Lifecycle:** Built cross-platform client with RevenueCat for paywall entitlements ($25 lifetime buyout), managing the complete production lifecycle from Figma UX design to App Store deployment for 1,000+ Micro 4/3 photography enthusiasts.

Software Developer & UX Designer — Aug 2024 - Sept 2026

Signsbeat Pte Ltd — *Singapore, Remote*

- **AI-Driven Clinical System & MCP Architecture:** Designed and deployed a multi-node AI agent pipeline (LangGraph) integrated into Web and App platforms to assist healthcare providers with patient care. Implemented Model Context

Protocol (MCP) strictly bound to internal enterprise knowledge, with multi-layered guardrails, context memory, and file processing to eliminate hallucinations and ensure precise clinical outputs.

- **Core Data Engine & Backend Standardization:** Re-engineered backend data processing architectures to ensure absolute data consistency across platforms. Centralized raw health metric calculations into the backend—resolving legacy client-side computation mismatches and unlocking critical enterprise sales deals that required strict data accuracy.
- **End-to-End Product Delivery:** Spearheaded the full product lifecycle of Yishii app, transitioning from UX research (Empathize to Ideate) to high-fidelity design and app development using React Native.
- **Scalable Architecture & Infrastructure:** Built Yishii from scratch, implementing a robust architecture featuring Zustand (State Management), Sentry (Error Tracking), and comprehensive testing suites (E2E & Unit Testing) to ensure crash-free sessions.
- **Hardware & Health Integration:** Engineered complex modules for Bluetooth connectivity and bi-directional data syncing with Apple HealthKit and Android Health Connect.
- **AI & Gamification Innovation:** Integrated an AI-powered nutritional analysis feature for food tracking and developed "Yishii Island," a gamified UX module that significantly increased daily active users (DAU).
- **DevOps & Future-Proofing:** Modernized the tech stack to a scalable framework for easier upgrades; single-handedly managed the end-to-end deployment and maintenance pipeline for App Store and Google Play.
- **Leadership & Documentation:** Authored the company's first comprehensive API documentation to ensure system scalability; mentored and led a growing engineering team following a successful seed funding round from Taiwan investors.

Creative Developer — June 2025 - Present

Freelance — *Malaysia*

- **Strategic Client Discovery:** Direct end-to-end consulting for diverse clients (e.g., professional emcees, educators), identifying brand positioning, target audiences, and core business objectives to align design strategy with **OKRs**.
- **Psychology-Driven Branding:** Defined visual identities by leveraging color and typography psychology to craft distinct brand voices, ensuring aesthetic consistency across all digital touchpoints.
- **Information Architecture & UX:** Engineered conversion-focused Information Architecture (IA) and wireframes, prioritizing user flow efficiency to meet specific project KPIs before high-fidelity execution.
- **High-End Creative Design:** Designed bespoke, high-performance websites with a focus on **custom interactive animations** and immersive storytelling, tailored for creative professionals.
- **Full-Stack Implementation:** Managed the entire development lifecycle from conceptual design to final deployment, ensuring responsive, cross-platform compatibility and optimal SEO performance.

PROJECTS

Personal Website — Portfolio | *Vanilla JS, ThreeJS, JSON* — Feb 2024 - Aug 2024

- **Pseudo-Dynamic Architecture:** Engineered a high-performance "pseudo-dynamic" website using **Vanilla JavaScript** and **JSON** as a lightweight data layer, achieving near-instant load times without the overhead of heavy frameworks.
- **Interactive 3D Experience:** Developed a custom project showcase using Three.js, implementing immersive 3D spatial navigation to enhance user storytelling and engagement.
- **Geospatial Travel Map:** Built a dynamic Interactive World Map from scratch to document global travels; integrated custom logic to parse JSON image metadata and render location-specific photo galleries dynamically.

- **Advanced Creative Coding:** Crafted complex CSS animations and native JS event listeners to ensure a seamless, fluid UX, demonstrating a deep mastery of core web technologies and browser APIs.

Real-time Second-Hand Goods Price Comparison System Jan 2024 - May 2024

| *.NET Core, React Native, Elasticsearch, Scrapy*

- **Full-Stack System Architecture:** Designed and implemented a cross-platform ecosystem, enabling real-time price comparisons across multiple second-hand marketplaces.
- **Scalable Data Engineering:** Developed an extensible web scraping engine using the **Scrapy framework**, featuring real-time incremental crawling to ensure data freshness and system reliability.
- **High-Performance Search:** Integrated Elasticsearch for efficient indexing and full-text search, significantly optimizing query response times for large-scale product datasets.
- **Robust Backend Development:** Built a server-side using C# .NET Core WebAPI and SQL Server, implementing features such as user subscriptions, automated push notifications, and international price normalization.
- **Multi-Platform Client:** Developed intuitive client applications using React Native and Vue.js, focusing on seamless data rendering and user-centric navigation.

TECHNICAL SKILLS

Language and Frameworks: React, React Native, Expo, TypeScript, JavaScript (ES6+), HTML5/CSS3, Three.js, C# (.NET Core).

AI and Architecture: LangGraph, Model Context Protocol (MCP), LLM Prompting, Zustand, React Query.

Tools and Infrastructure: Figma, Git/GitHub, Sentry, RevenueCat, Postman, App Store Connect, Google Play Console.

Communication Language: English (Fluent), Mandarin (Native)